Terrestrial protected areas stabilize the inland habitats of diadromous fishes under climate change

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Terrestrial protected areas stabilize the inland habitats of diadromous fishes under climate change

Abstract: Diadromous fishes (D-Fishes) represent a unique group of species that migrate between freshwater and marine environments. They provide a major protein source for humankind and sustain important fishery-based economies. However, the global distribution and conservation status of D-Fishes remain poorly documented. Investigating their habitat shifts and evaluating the effectiveness of existing protected areas are thus essential. Here, we employed the Maximum Entropy algorithm within Species Distribution Models to model the freshwater habitats of global D-fishes from 1970 to 2020 and projected their distributions under future global change scenarios. We show that the highest richness of D-fishes occurs primarily in the coastal regions of Oceania, as well as along Southeast Asia and eastern India coasts. Amphidromous and catadromous fishes are predominantly distributed in Oceania, whereas anadromous fishes are primarily found in North America. Losses in species richness are predicted to expand across large areas under future (2090) climate change scenario SSP5-8.5. Based on model outputs, we further assessed the effectiveness of protected areas. Furthermore, protected terrestrial areas increase the proportion of stable habitats by 5% to 27%, although their protective efficacy diminishes as climate change intensifies. midwestern region of NA emerging as a priority area for the establishment of new protected areas. Therefore, increased attention to the conservation of freshwater habitats for D-fishes is crucial.

Keywords: Diadromous fishes; Freshwater habitats; Climate change; Richness patterns; Protected areas

Introduction

Diadromous fishes (D-Fishes) migrate between marine and freshwater environments, with many species known to undertake long-distance (>1000 km) migrations (McDowall, 1988; Elliott & Hemingway, 2008). They play pivotal ecological roles, acting as vectors for nutrient transport and as integral components of aquatic food webs (Poulet et al., 2022; Waldman & Quinn, 2022). D-Fishes also represent a significant source of protein for humans (Kuroki et al., 2014; Dekker, 2019), a foundation for fishery economies (Marta et al., 2001; Dekker, 2003), a medium for recreational

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activities (Fortin et al., 1990) as well as being embedded in cultural beliefs (Rutherford, 1971; Clemens et al., 2017). Yet, D-Fishes populations have experienced significant declines over the last century (Lucas & Baras, 2008; Waldman & Quinn, 2022), particularly in North America and Europe (Limburg et al., 2003; Buysse et al., 2008). To restore the population and diversity of D-Fishes, various measures have been implemented, including fishing restrictions (Richards & Rago, 1999), artificial breeding and stock enhancement (Lichatowich, 1999), water pollution treatment (Sheppard et al., 2014), dam removal (Bednarek, 2001) as well as the construction of fish passage facilities (Croze et al., 2008; Franklin et al., 2012). Despite some local successes (Meadows, 2001; Merg et al., 2020), most of these measures have not been sufficient to reverse the overall decline in D-Fish populations (Lichatowich, 1999; Bunt et al., 2012; Waldman & Quinn, 2022, Dambrine et al., 2023). Due to their long-distance migrations, D-Fishes are facing multiple threats, rendering them more sensitive to anthropic disturbances than non-migratory fishes (Dambrine et al., 2023), and making their conservation particularly challenging (Verhelst et al., 2021).

Transnational conservation efforts, are recognized as an effective strategy for the protection of D-Fishes (Verhelst et al., 2021; Dambrine et al., 2023), with previous US-Canada collaborative initiatives demonstrating measurable success in restoring Fraser river salmon populations (Roos, 1991). For instance, Lassalle et al. (2008) developed habitat models across continents for 22 species of D-Fishes to assess the potential impacts of global warming, while Barbarossa et al. (2020) evaluated the global connectivity of terrestrial habitats for these species. These studies have laid a theoretical foundation for furthering international collaboration in the conservation of D-Fishes. Moreover, the establishment of protected areas for D-Fishes is also considered beneficial (Saunders et al., 2002). However, to date, global-scale research on the distribution of freshwater habitats used by D-Fishes and their projected changes under future scenarios remains unreported, and the effectiveness of existing terrestrial protected areas in conserving these habitats has yet to be evaluated.

To address the existing gap in this field, we utilized the species names of D-Fishes from Barbarossa et al. (2020) as a basis for geographic information collection,

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screening, and filtering. Subsequently, we employed the MaxEnt (Maximum Entropy) algorithm (Phillips et al., 2006) within Species Distribution Models, considering Temperature Seasonality, Precipitation Seasonality, Dam Density and Human Footprint as predictor variables, to develop global freshwater habitat models for D-Fishes. These models were used to predict current and future species richness distribution patterns and their projected changes. Furthermore, we integrated the recent status of global terrestrial protected areas (WDPA; Version Jan. 2025) distribution data to evaluate the effectiveness of these areas in conserving the freshwater habitats of D-Fishes.

Methods

D-Fishes occurrences, assessment, and classifications

The list of D-Fishes was extracted from the study by Barbarossa et al. (2020), and a total of 490 D-Fishes were determined. The occurrences of these D-Fishes in this study were primarily obtained from three sources Global Biodiversity Information Facility (GBIF) (https://www.gbif.org/), and FishNet2 (http://www.fishnet2.net/). Some supplementary data were s retrieved from comprehensive regional databases covering Asia and South America (Jézéquel et al., 2020; Tonella et al., 2023; Du et al., 2024); In total, 6,816,489 occurrence records were compiled using 490 D-Fishes species names for subsequent processing. For the preliminary dataset, we first checked for taxonomic synonyms and naming errors and verified the spatial accuracy of the all records (Da Mata et al., 2017). To mitigate the effects of spatial autocorrelation, we applied spatial thinning using the "spThin" R package (Aiello-Lammens et al., 2015), retaining only one record per 2.5 arcmin grid cell (Boria et al., 2014; Steen et al., 2021). After processing, 56,881 occurrence records were retained for modeling. To incorporate biogeographically realistic dispersal limitations (as D-Fishes cannot cross continental barriers), the globe was divided into seven regions, and species distribution models were constructed separately for each region. The delineation of these study regions followed the basin divisions from GRNWRZ V2.0, which accounts for both basin area and river network connectivity within each water resource zone (Yan et al., 2022).

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Occurrence records were assigned to their corresponding regions and used accordingly for modeling.

Models for 275 of 490 D-Fishes species with more than 5 occurrences were successfully developed. These species were cross-referenced with the IUCN Red List to retrieve global conservation assessments. Given that different migratory types of D-Fishes may exhibit distinct distribution patterns and characteristics, we also determined the migratory type (amphidromous, anadromous, catadromous) for each 275 species using targeted searches in Google Scholar (https://scholar.google.com; retrieved in December 2024). Species with uncertain, controversial, or undefined migratory types were classified as "other diadromous fish".

Predictor variables selection

The selection of predictors can significantly influence the quantification of ecological niches, thereby affecting the temporal and spatial transferability of individual SDMs (Townsend Peterson et al., 2007). To ensure the reliability of SDMs, predictor variables must be ecologically relevant to the modeled species (Sillero et al., 2021). We selected Temperature Seasonality (BIO4), Precipitation Seasonality (BIO15), Dam Density (DD), and Human Footprint (HFP) as predictor variables. All variables were set or resampled to match the resolution of the occurrence data (2.5 arcmin). BIO4 and BIO15, two climate variables sourced from the WorldClim database (http://www.worldclim.org/; Fick & Hijmans, 2017), are frequently used in modeling species distributions under climate change scenarios (Guisan et al., 2017; Mod et al., 2016). HFP (https://zenodo.org/records/4935069; Venter et al., 2016) represents the human footprint in 2009, calculated by overlaying pressures such as built environments, population density, electric infrastructure, crop lands, pasture lands, roads, railways, and navigable waterways. DD is a global dataset of dam density (values ranging from 0 to 1), developed by Vörösmarty et al. (2010) (https://www.riverthreat.net/data.html). The selection of these variables was guided by ecological rationale D-Fishes richness shows strong seasonal variation, which is closely linked to their seasonal migratory

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activity (Vedra et al., 2013; Chan et al., 2024); thus, BIO4 and BIO15 were used to capture climatic drivers of migration. Migratory behaviors of D-Fishes often conflict with various human uses of aquatic environments (Verhelst et al., 2021; Dambrine et al., 2023), and HFP was included to anthropogenic pressures. DD is a key factor in the decline of D-Fishes populations (Merg et al., 2020), as dams obstruct migratory routes, fragment habitats, and contribute to population declines (Waldman & Quinn, 2022). Including DD as a predictor thus helps account for the effects of habitat fragmentation and disrupted migratory connectivity.

To avoid overfitting and reduce the influence of multicollinearity (De Marco & Nóbrega, 2018), we conducted multicollinearity tests for BIO4, BIO15, HFP, and DD across the seven study regions. All variables had Variance Inflation Factor (VIF) values below 5 (Supporting Information I, Table 1), and all pairwise Pearson correlation coefficients had absolute values below 0.8 (Graham, 2003; Yu et al., 2013; Liu et al., 2019; Supporting Information I, Figure 1), indicating weak multicollinearity. Maintaining low multicollinearity is expected to improve SDM transferability and enhance their ability to predict species' adaptability to novel environmental conditions (Biancolini et al., 2024).

MaxEnt Modeling and Prediction

This study employed the MaxEnt algorithm (Phillips et al., 2006) to construct species distribution models (SDMs). Models were fitted using the default settings of the "biomod2" package (Thuiller et al., 2016), within the open-source statistical software R 4.3.1. Model performances were evaluated using the AUC (Swets, 1988) and TSS (Allouche et al., 2006). Subsequently, we applied repeated split-sample and cross-validation procedures to assess model performance, retaining only models with AUC ≥ 0.75 for the construction of an AUC-weighted ensemble model (Araújo & New, 2007; Buisson et al., 2010). Specifically, 70% of the presence records and an equal number of randomly generated pseudo-absence records were randomly selected for training, while the remaining 30% were used for validation (Bai et al., 2024). This process was

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repeated 10 times to minimize the bias associated with random data partitioning and to increase model robustness.

We initially modeled the distribution of D-Fishes for the historical period (1970–2020), in which 1970 is the common reference year of the Living Planet Index (LPI) by WWF's report (Almond et al., 2020), and is a close proximity to the LPI for migratory fish (Deinet et al., 2020). Future projections were then generated for 2050 (averaged over 2041–2060) and 2090 (averaged over 2081–2100), keeping non-climatic variables (HFP and DD) constant. HFP was held constant due to the absence of reliable future data. Although some projections of future dam construction exist (https://www.globaldamwatch.org/fhred), these datasets do not include many small dams and are incomplete for certain countries (Barbarossa et al., 2020). Moreover, dam removal is actively ongoing, particularly in North America and Europe, where the number of removed dams may exceed newly constructed ones (https://www.americanrivers.org; https://damremoval.eu; Habel et al., 2020). Therefore, future DD values were considered too uncertain to incorporate and were kept constant across time periods. The use of static non-climate variables in predicting the impacts of climate change is widely accepted (Peterson et al., 2002), and previous studies have shown that models including both static and dynamic variables perform better than those that omit or exclude static variables (Iverson & Prasad, 1998; Stanton et al., 2012). To reduce uncertainty associated with future climate projections, we selected meteorological products from three widely used global circulation models (ACCESS-CM2, CanESM5, and MIROC-ES2L) (Supporting Information I, Table 2), and averaged projected distributions of D-Fishes richness across these models (Hole et al., 2009; Mi et al., 2023). Future scenarios were based on the Shared Socioeconomic Pathways (SSPs), developed for the IPCC Sixth Assessment Report (AR6), which reflect different levels of global carbon emissions and socioeconomic development (O'Neill et al., 2017). We selected two contrasting SSP scenarios: SSP1-2.6 (a low-emission, mitigation-oriented scenario) and SSP5-8.5 (a high-emission, business-as-usual scenario), to predict future distributions of D-Fish habitats.

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Identification and evaluation of habitat protection based on WDPA

To assess habitat changes under climate change based on SDMs for 275 D-Fishes, a habitat suitability (HS) threshold was established for each model. This threshold was set at 0.7 times the maximum HS value observed during the modeling period. Grid cells with HS values above this threshold were classified as habitats, while those below the threshold were classified as non-habitats. This threshold remained constant for future periods.

WDPA, the most comprehensive global database in terrestrial and marine protected areas, has been widely used in biogeography and conservation biology studies at both national and global scales (Li et al., 2024; Mi et al., 2023). Using the "wdpar" package (Hanson, 2022) in R, we obtained data on Protected Areas (Version Jan.2025), cleaned the data by country according to the developer's recommendations, and merged them into a global Protected Areas dataset. The data processing steps were as follows: First, ISO codes were retrieved for all countries to obtain the corresponding Protected Areas shapefiles; Second, Protected Areas that have not been formally implemented or are considered to have limited conservation value, such as those with a status of proposed, not reported, or designated as UNESCO MAB biosphere reserves were excluded.; Third, point-based Protected Areas were converted into circular polygons based on their reported areas; Fourth, the Protected Areas shapefiles were projected onto the WGS84 coordinate system. Then, shapefiles for all countries were merged into a global Protected Areas distribution map. Finally, we divided the habitats of the 275 D-Fishes species into those inside and outside Protected Areas. This allowed us to assess habitat changes under climate change separately for protected and unprotected regions, and to evaluate the conservation effectiveness of these PAs. Based on the WDPA Protected Areas distribution and projected changes in D-Fishes richness under climate change, we calculated the PU using the following formula:

$$RLOP = (SR_{i,j} - SR_{2000}) \times (PM - 1)$$

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$$PU = f(RLOP)$$

Where, $RLOP$ represents the SR loss of D-Fishes outside protected areas; SR_{2000} represents the species richness of D-Fishes during 1970–2020, and $SR_{i,j}$ represents the species richness of D-Fishes during period j under future climate scenario i; PM is a binary map of protected area distribution, with 1 indicating the presence of a protected area and 0 indicating its absence; $f(x)$ is a classification function such that: When $RLOP < 1$, $PU = 0$; When $1 \le RLOP < 2$, $PU = 1$; When $2 \le RLOP < 5$, $PU = 2$; When $10 \le RLOP < 20$, $PU = 4$.

Results

Species composition, models performance, and predictors contribution

A total of 275 D-Fishes were successfully modeled, encompassing 25 orders. The three most speciose orders were Anguilliformes (90 species, 32.73%), Perciformes (30 species, 10.91%), and Salmoniformes (27 species, 9.82%). These species were distributed across 63 families, with the most species-rich families being Gobiidae (67 species, 24.36%), Salmonidae (27 species, 9.82%), and Eleotridae (22 species, 8.00%). According to the International Union for Conservation of Nature (IUCN) Red List (https://www.iucnredlist.org/; assessed in December 2024), 77.82% of D-Fishes were categorized as Least Concern, while only 3.64% (10 species) were classified as Endangered or Critically Endangered. Among these D-Fishes, 63.64% of the species had unknown population trends, 21.82% were considered stable, 12.73% were decreasing, and only 1.10% (3 species) were reported to have increasing trends. Through a review of 170 literature sources, the migratory types of the 275 D-Fishes were categorized as follows: amphidromous (32.00%; species that migrate between freshwater and ocean, but not necessarily for reproduction, often for feeding or growth), anadromous (23.64%; species born in freshwater, migrating to the ocean to grow, and returning to freshwater to reproduce), catadromous (12.73%; species born in the ocean, migrating to freshwater to grow, and returning to the ocean to reproduce) as well as

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others (31.64%) (Supporting Information II).

Model performance was excellent, with median AUC (Area Under the Curve of Receiver Operating Characteristic) and TSS (True Skill Statistics) values for all models across the seven study regions exceeding 0.92 and 0.75, respectively (Figure 1d). Across all models, regardless of study region, the median AUC and its SD (standard deviation) were 0.94 and 0.08, respectively, while the median TSS and its SD were 0.81 and 0.14, respectively. Among the four modeling variables, Temperature Seasonality (BIO4) exhibited the highest relative contribution, whereas Human Footprint (HFP) showed the lowest (Figure 1e). The average relative contributions of Temperature Seasonality (BIO4), Precipitation Seasonality (BIO15), Dam Density (DD), and Human Footprint (HFP) were 49.90%, 19.14%, 24.74%, and 6.22%, respectively.

Current potential species richness distribution

A total of 429 models were constructed for D-Fishes across seven study regions, the proportion of D-Fishes models for different migratory types varies across different study regions (Figure 1c). The predicted global potential species distribution of D-Fishes, which matched the observed occurrence data (Figure 1b), was illustrated in Figure 1a. Our results indicated that the region with the highest species richness (SR) ($40 \le SR < 60$) was located exclusively along the coastal areas of Oceania (OC), covering an area of 7.30×10^4 km^2. The regions with the second-highest SR ($30 \le SR < 40$) were predominantly distributed along the coastal areas of Southeast Asia (SA) and southern India, encompassing an area of 1.71×10^6 km^2, which accounts for 69.60% of the total area of these secondary high-richness regions. The distribution of D-Fishes was widespread across the study regions (except for ASa), with their presence ($SR \ge 5$) covering over 60.00% of the total area in each respective region.

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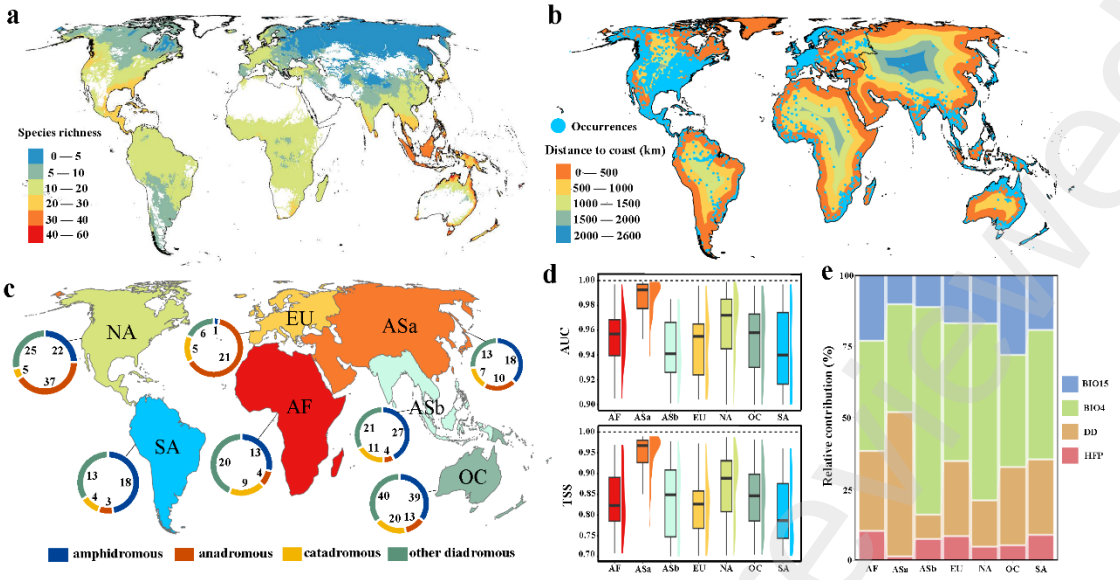


Figure 1 Global potential species richness distribution patterns and model overview of diadromous fishes (D-Fishes). **a** Global potential species richness distribution in 1970—2020. **b** Processed occurrence records in 1970—2020 used for modeling. **c** Study regions and number of D-Fishes models for different migratory types constructed per region. **d** AUC (Area Under the Curve of Receiver Operating Characteristic) and TSS (True Skill Statistics) values of models across the seven study regions. **e** Average contribution of modeling variables across the seven study regions.

Future species richness change under climate change

The future species richness (SR) of D-Fishes was compared with the historical period (1970—2020), and the resulting distribution of species richness change (SRC) under climate change was illustrated in Figure 2. A slight decline in SR (Represent by CM1: $-5 \leqslant SRC < -1$; the same as followed) was predominantly observed in the South America (SA) region, with little variation in area across different scenarios and time periods, averaging 5.00×10^6 km^2. Under the SSP5-8.5 scenario, CM1 exhibited an increase across all study regions (comparing the 2090 period to the 2050 period), with the most pronounced increases in the NA, EU, and ASa regions, showing net area increases of 3.47×10^6 km^2, 1.00×10^6 km^2, and 2.02×10^6 km^2, respectively. A significant decline in SR (CM2: $SRC < -5$) was almost exclusively observed in the NA

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region under the scenario of SSP5-8.5 in 2090, covering an area of 4.27 × 10^5 km^2, while in other regions or periods, the CM2 area was less than 7.00 × 10^3 km^2. A slight increase in SR (CM3: 1 ≤ SRC < 5) was mainly distributed in the Africa (AF) and NA regions, with the former region showing little variation in CM3 area across different scenarios and time nodes (averaging 7.23 × 10^6 km^2) and the latter region experiencing a sharp decline under the scenario of SSP5-8.5 in 2090 (7.54 × 10^6 km^2, compared to an average of 1.14 × 10^7 km^2 in other periods). CM3 increased in other study regions, with the largest increase observed in ASa under the scenario of SSP5-8.5 in 2090, which saw an increase of 9.37 × 10^5 km^2 compared to the 2050 period. A significant increase in SR (CM4: $SRC \geq 5$) was primarily observed in the OC region, with large areas across different scenarios and time nodes, averaging 4.48 × 10^5 km^2, and only NA reached a similar magnitude under the scenario of SSP5-8.5 in 2090 (3.88 × 10^5 km^2).

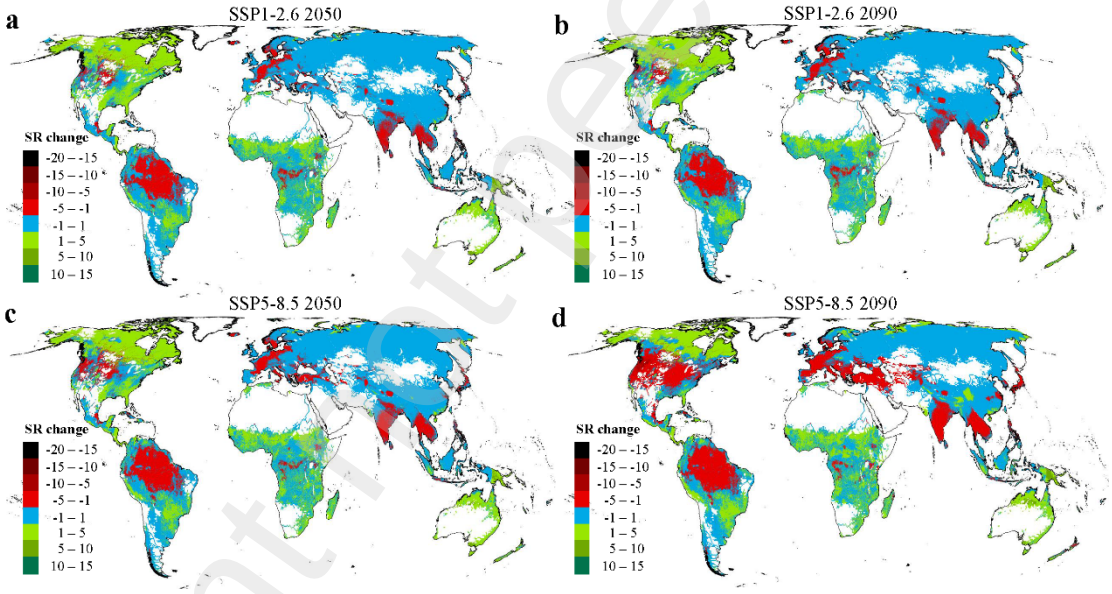


Figure 2 Global distribution of future changes in the species richness of D-fishes.

Future trends of species richness change for different groups of D-Fishes

The bivariate distributions of species richness in the historical period (2010) and changes in species richness under the SSP5-8.5 scenario for the 2090 period are shown in Figure 3 for amphidromous (AMF), anadromous (ANF), catadromous (CAF), and other diadromous (ODF) fishes. Corresponding distributions for SSP1-2.6 scenario in

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the 2050 and 2090 period, as well as SSP5-8.5 scenario in the 2050 period, are presented in Supporting Information I Figures 2, 3, and 4, respectively.

AMF, ANF, and ODF are primarily distributed in the OC region, while ANF are mainly concentrated in the NA region. In terms of high-risk areas (AT1: Lowest and decreased richness), ANF exhibits the smallest area, which further declines under future scenarios, with an 85.2% reduction from the 2050 period to the 2090 period under the SSP5-8.5 scenario. Similarly, high-risk areas for AMF, ANF, and ODF are predominantly located in the NA region, reaching their maximum extents in the 2090 period under SSP5-8.5, with areas of 2.31×10^6 km², 1.11×10^6 km², and 2.78×10^6 km², respectively. Regarding low-risk areas (AT9: Highest and increased richness), AMF, CAF, and ODF exhibit only small-scale distributions in the OC region, with relatively stable area sizes across the four time periods, averaging 8.40×10^4 km², 1.05×10^4 km², and 1.32×10^5 km², respectively. In contrast, the low-risk areas for ANF are confined to the NA region, and their extent under SSP5-8.5 is notably smaller than under SSP1-2.6, averaging only 40.1% of the latter.

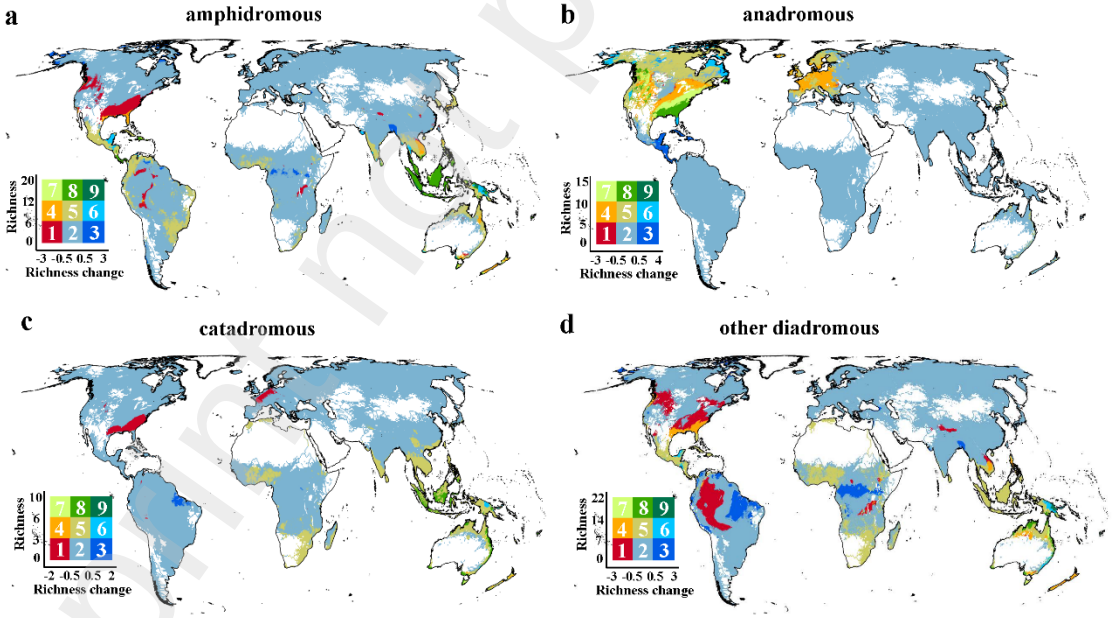


Figure 3 Bivariate plots of species richness distribution in the historical period (1970—2020) and species richness changes under the scenario of SSP5-8.5 in 2090 period. Panels (**a**, **b**, **c**, **d**) correspond to different migratory types: amphidromous, anadromous, catadromous, and other diadromous species, respectively. "1" represents Area of Type 1, defined as the highest risk; "9" are similarly defined as the lowest risk; Note that the same colors in the legends correspond to different value ranges in these plots.

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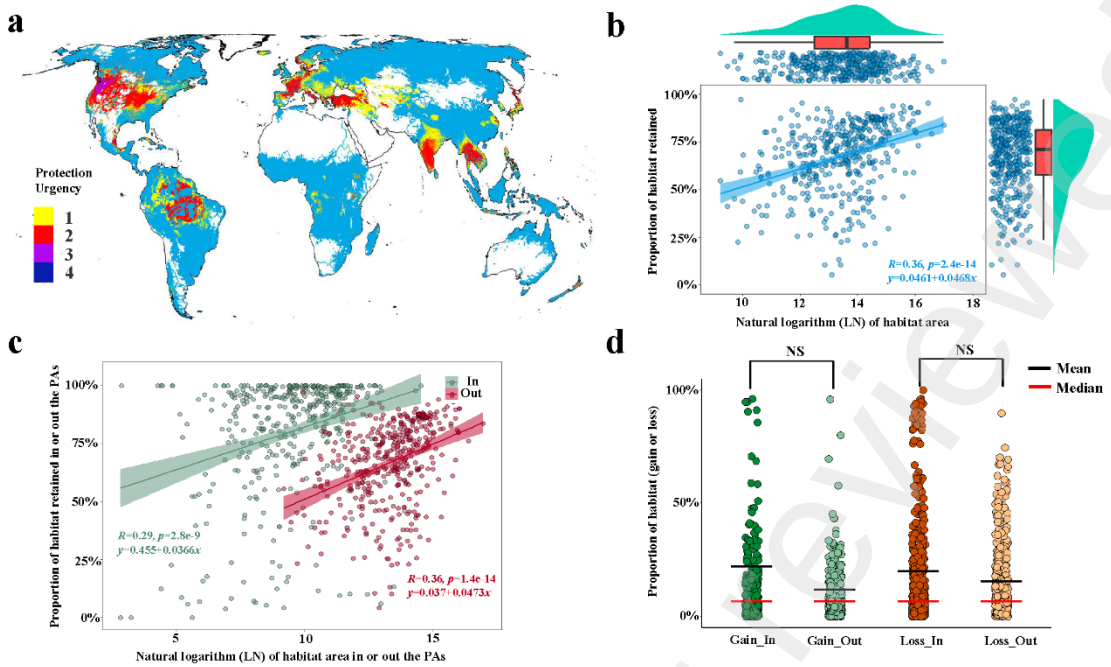
Benefits of WDPA protected areas and protection urgency

The conservation status and effectiveness of protected areas for D-Fishes under the scenario of SSP5-8.5 scenario in 2090 period were illustrated in Figure 4 (corresponding figures for scenarios of SSP1-2.6 scenario in 2050 period, SSP1-2.6 scenario in 2090 period, and SSP5-8.5 scenario in 2050 period were presented in Supporting Information I, Figures 5, 6, and 7). Under climate change, the proportion of D-Fishes habitats that remain unchanged (Retain Ratio; RR) was positively correlated with the total habitat area (Figure 4b; Supporting Information I, Figures 5b, 6b, 7b). However, patterns differed inside protected areas. Although the total habitat area of D-Fishes habitats within protected areas was smaller than outside, the RR was consistently higher within protected areas (Figure 4c; Supporting Information I, Figures 5c, 6c, 7c). With intensifying climate change (e.g., under the SSP5-8.5 scenario compared to SSP1-2.6), the effectiveness of habitat protection declines across most study regions (Table 1). Nevertheless, the presence of protected areas can increase the proportion of stable habitats by 5% to 27%. Additionally, the extent of habitat gained and loss for D-Fishes did not show a significant relationship with whether the habitats were located inside or outside protected areas.

Considering both habitat and SR loss outside protected areas, the Protection Urgency (PU) for D-Fishes appeared high across most global regions, except in the AF and OC regions (Figure 1a). Areas with the highest PU (PU > 2) were almost exclusively located in the NA region under the scenario of SSP5-8.5 scenario in 2090 period, covering an area of 4.21×10^5 km^2. In contrast, the area with PU > 2 in all other periods or regions was less than 7.00×10^3 km^2 (Figure 4a; Supporting Information I, Figures 5a, 6a, 7a).

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346

347 **Figure 4** Conservation status of D-Fishes under the scenario of SSP5-8.5 in 2090. **a**
348 Conservation gaps and urgency of D-Fishes. **b** Relationship between the total habitat
349 area in 2010 and the proportion of habitat retained under climate change. **c** Relationship
350 between the total habitat area inside (In) and outside (Out) protected areas (PAs) and
351 the proportion of habitat retained under climate change. **d** Proportion of habitat gained
352 and loss under climate change, categorized by habitats inside (In) and outside (Out)
353 PAs. NS means Not Significant; each scatter point represents habitat model of a
354 diadromous fish.

355

356 **Table 1** Difference of RR between inside and outside protected areas for seven study
357 regions; RR refers to the proportion of D-Fishes habitats that remain unchanged.

	SSP1-2.6 2050	SSP1-2.6 2090	SSP5-8.5 2050	SSP5-8.5 2090
AF	5.83%	5.28%	6.07%	6.91%
AS1	17.54%	17.35%	15.03%	11.89%
AS2	4.58%	4.16%	4.33%	5.31%
EU	26.00%	26.53%	25.74%	24.63%
NA	6.44%	7.24%	11.05%	13.15%
OC	18.70%	19.24%	18.89%	18.51%

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SA	14.74%	14.55%	13.50%	10.27%

Discussion

By investigating and processing global occurrence data of D-Fishes from 1970–2020, we successfully developed SDMs for more than half (275/490) of the D-Fishes species. Utilizing these models, we revealed the distribution and changes in their freshwater habitats and evaluated the effectiveness of terrestrial protected areas in mitigating the impacts of climate change. Our research provides valuable insights into the conservation of D-Fishes freshwater habitats and offers effective references for the planning and development of global terrestrial protected areas.

The taxonomic survey of modeled D-Fishes reveals a pronounced clustering of taxa, with the orders Anguilliformes (32.73%), Perciformes (10.91%), and Salmoniformes (9.82%), along with dominant families such as Gobiidae (24.36%) and Salmonidae (9.82%), constituting the majority of this group. The high degree of taxonomic concentration reflects the adaptive advantages of specific taxa to the diadromous life history, further illustrating the relatively unified evolutionary trends of D-Fishes. For instance, in order to adapt to prolonged migrations and overcome obstacles along migratory routes, D-Fishes have evolved larger body sizes compared to typical freshwater fishes (Hendry et al., 2004; Bloom et al., 2018). Additionally, to reduce early-life mortality and better match the new environments (Chambers & Trippel, 1997; Augspurger et al., 2017), D-Fish produce larger eggs than typical freshwater fish (Augspurger & Closs, 2019), which are believed to better adapt to harsh environments (Jones et al., 2017). Unfortunately, these adaptive evolutionary traits are insufficient to cope with the multiple pressures and threats of the Anthropocene (Waldman & Quinn, 2022), as both the habitats and population sizes of D-Fishes have experienced significant declines (Limburg et al., 2003; Hall et al., 2012). Although 77.82% of modeled D-Fishes were assessed as Least Concern on the IUCN Red List, with only 10 species (3.64%) classified as Endangered or Critically Endangered, 12.73% and 63.64% of the populations are reported to be declining or have uncertain trends,

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386 respectively. This disproportionate data gap suggests a potential underestimation of the
387 actual risks faced by D-Fishes in conservation assessments, and an underestimation of
388 the impacts upon population declines. Though not yet extinct, many D-Fishes have seen
389 their numbers drastically reduced to levels where they can no longer fulfill their original
390 ecological roles, earning them the designation of "ghost species" (Waldman, 2013).

391 In the habitat models for D-Fishes, the importance of climatic factors represented
392 by BIO4 and BIO15 significantly outweighs anthropogenic factors represented by DD
393 and HFP (Figure 1e). This result indicates that the distribution of D-Fishes habitats is
394 more dependent on climate or climate-driven seasonal variations, which aligns with
395 existing knowledge (Shiao et al., 2003; Dinh et al., 2017; Itakura & Wakiya, 2020). For
396 instance, the spawning migrations of anadromous fish (e.g., salmonids) are tightly
397 synchronized with specific water temperature thresholds and precipitation-driven flow
398 changes (Jonsson & Jonsson, 2009), while the larval dispersal of amphidromous fish
399 (e.g., gobies) may rely on estuarine salinity gradients regulated by seasonal ocean
400 currents (Whitfield et al., 2023). The high contribution of dam density underscores the
401 core threat posed by migratory barriers to D-Fishes (Barbarossa et al., 2020; Waldman
402 & Quinn, 2022), a threat often considered more severe than that faced by typical
403 freshwater fishes (Cote et al., 2009). Dams can sometimes directly cause mortality in
404 D-Fishes; for example, female American eels traversing dams on the St. Lawrence
405 River experience survival rates of less than 2% due to turbine injuries (MacGregor et
406 al., 2015). Although the importance of the human footprint is relatively low, its actual
407 impact on D-Fishes may be underestimated by the models, as human activities
408 represented by the HFP (e.g., urbanization, agriculture, and transportation pressures)
409 can influence their distribution through indirect pathways (Su et al., 2021).

410 The regions with the highest species richness of D-Fishes are exclusively located
411 in the coastal areas of OC, while the second-highest richness areas are concentrated in
412 ASa and the eastern coastal regions of India (ASb) (Figure 1a). Firstly, the complex
413 estuarine and marine environments in these regions may provide diverse spawning and
414 nursery grounds for D-Fishes (Elliott & Hemingway, 2008). Secondly, island
415 ecosystems, such as the coral reef-mangrove complexes in OC, likely support a high

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proportion of endemic species differentiation due to the intricate structures of freshwater-marine habitat ecotones (Nagelkerken et al., 2008). Given their complex life cycles, D-Fishes are particularly vulnerable to the impacts of climate change (Barber-O'Malley et al., 2022). The research results indicate that the area of species richness loss for D-Fishes gradually expands under climate change, reaching its peak under the scenario of SSP5-8.5 in 2090 period (Figure 2). While the OC region holds the highest species richness, it also exhibits the greatest increase in richness. Notably, in the NA region, both the areas of significant richness increase and decrease peaked under the scenario of SSP585 in 2090 period. The increases and decreases in D-Fishes richness in NA are concentrated in high-latitude and low-latitude areas, respectively, revealing a phenomenon of richness centers shifting toward higher latitudes.

Amphidromous fishes, as a key group in tropical-subtropical estuarine ecosystems, exhibit the highest species richness in the OC region. However, the area of their lowest-risk zones during the SSP5-8.5 scenario in 2090 period decreased by 27.6% compared to the SSP1-2.6 scenario (Figure 3a). This reduction may be attributed to the warming and salinity changes in seawater under high-emission scenarios, which could disrupt the freshwater-marine habitat gradients essential for their larval dispersal (Iguchi & Takeshima, 2011; McCormick & Regish, 2018). The expansion of high-risk zones in the NA region may be linked to the failure of larval migration due to inland river temperatures exceeding critical thresholds (Murase et al., 2019). Anadromous fishes, as representative cold-water species, experienced a dramatic expansion of medium-risk zones in the NA and EU regions under the scenario of SSP5-8.5 (increasing by 2.13 times and 4.90 times, respectively, from 2050 period to 2090 period; Figure 3b). This expansion may result from accelerated winter snowmelt, rising summer temperatures, and reduced river flows in high-latitude regions, which impact their spawning migrations (Mote, 2003). Catadromous fishes, whose habitats in the OC and ASb regions are mostly in low-risk zones, saw a sudden and large-scale emergence of high-risk zones in the southeastern coastal NA and western coastal EU regions under the scenario of SSP5-8.5 in 2090 period (Figure 3c). Under the scenario of SSP5-8.5, the intensification of the Gulf Stream in southeastern NA and the variability of the North

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446 Atlantic Current in the western EU may lead to abnormally high coastal seawater
447 temperatures (e.g., > 30 °C in summer) and disrupted salinity gradients (Muller-Karger
448 et al., 2015; Hand et al., 2019; McCormick & Regish, 2018). These changes could
449 directly inhibit spawning behaviors in adults (e.g., temperature constraints in eel
450 spawning areas; Tsukamoto, 1992; Aarestrup et al., 2009) and affect the larval drift,
451 thereby affecting their physiological behaviors when entering freshwater rivers or
452 eliminating rivers from the utilization by the early life phases.

453 The stability of D-Fishes habitats is positively correlated with the size of the
454 habitat area (Figure 4b, c), a pattern that also applies to typical freshwater fishes (Wood
455 et al., 2014; Greig et al., 2022) or marine fishes (Mellin et al., 2010). Although the
456 WDPA primarily focuses on terrestrial biodiversity rather than fish, it is evident that
457 protected areas enhance habitat stability (smaller habitat areas within protected areas
458 exhibit higher stability than bigger habitat areas outside protected areas; Figure 4c).
459 This phenomenon may arise from two factors: First, the selection of protected areas
460 often prioritizes key ecological nodes (e.g., migratory corridors and spawning grounds)
461 (Hofman et al., 2018; Liang et al., 2018), which may be less susceptible to climate
462 change disturbances; Second, protected areas effectively reduce human activities
463 (Geldmann et al., 2019), thereby enhancing habitat stability. However, as climate
464 change intensifies, the stabilizing effect of protected areas on habitats diminishes (Table
465 1), consistent with existing knowledge (Asamoah et al., 2021). Notably, the proportions
466 of habitat gain and loss for D-Fishes show no significant relationship with whether the
467 habitats are inside or outside protected areas (Figure 4d; Supporting Information I,
468 Figures 5d, 6d, 7d). This may be because terrestrial protected areas, when addressing
469 D-Fishes in rivers, focus more on preserving existing high-quality habitats rather than
470 restoring or improving habitat quality.

471 Due to objective limitations, our study has some shortcomings. First, due to the
472 lack of occurrences, only 56.1% of D-Fishes were considered in this study, and the
473 marine habitat portions of these species were not included. We agree with Ouellet et al.
474 (2022) and call for future research to consider both freshwater and marine habitats of
475 D-Fishes to further explore the patterns of habitat changes. Second, although DD was

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used as one of the modeling factors, our SDMs did not fully account for the dispersal capabilities of D-Fishes. The pace of climate change may exceed the ability of many species to track suitable habitats, especially in regions with severely fragmented river networks (Devictor et al., 2008; Loarie et al., 2009; Bai et al., 2024; Bai et al., 2025), and particularly for D-Fishes, which are more constrained by fragmentation than typical freshwater fish (Cote et al., 2009). In fact, the "megaSDM" (Version 2.0) developed by Shipley et al. (2022) can incorporate species-specific dispersal rates into the SDMs framework to account for dispersal capabilities, but the lack of reliable methods for calculating this parameter led us to abandon it. Third, this study only used the MaxEnt algorithm among SDMs; incorporating more algorithms could reduce the uncertainty of a single algorithm and provide more robust and reliable predictions (Grenouillet et al., 2011). Finally, treating human activities as static factors may underestimate their impact. Human activities are unlikely to remain static over the next 100 years, and for D-Fishes, factors such as water pollution and habitat degradation caused by human activities may pose greater threats than climate change (Dudgeon et al., 2006; Reid et al., 2019).

Data and Code Availability Statement

Supporting Information I (including seven figures and two tables) has been submitted to ***Biological Conservation*** along with this Manuscript. All information about diadromous fish (including taxonomic ranks, IUCN assessment, migration types and references) has been listed in Supporting Information II. Supporting Information II and R code used to conduct this study has been submitted to Dryad (https://datadryad.org/share/UOkKCH1KnjtG4yudoWTvq0MEMFNaezKrlc6MVCWBWXk). BIO4 (temperature seasonality) and BIO15 (precipitation seasonality) data are available at https://www.worldclim.org/data/worldclim21.html. DD (Dam Density) data is available at https://www.riverthreat.net/data.html. HFP (Human Footprint) data is available at https://zenodo.org/records/4935069. terrestrial Protected areas Data is available at https://www.protectedplanet.net/en/thematic-areas/wdpa?tab=WDPA. The

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delineation of the study regions from GRNWRZ Version 2.0 is available at https://figshare.com/articles/dataset/A_data_set_of_global_river_networks_and_corresponding_water_resources_zones_divisions_V2_0/17430749.

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